Pakistani Politician Image Classification Using Convolutional Neural Networks

Saim*, Abdullah†, Jawad‡
Department of Artificial Intelligence and Data Science, FAST-NUCES, Faisalabad Campus, Faisalabad, Pakistan
*f230018@cfd.nu.edu.pk †f230014@cfd.nu.edu.pk ‡f230028@cfd.nu.edu.pk

Supervised by:
Sir Asif Ameer, Omer Farooq Khan, Aun Ali, Ahsan Butt
Department of Artificial Intelligence and Data Science, FAST-NUCES, Faisalabad Campus, Faisalabad, Pakistan

***Abstract*—Automated facial recognition of public figures in domain-specific contexts poses unique challenges due to variations in pose, illumination, age, and image quality. This paper presents a deep-learning-based image classification system capable of recognising 16 prominent Pakistani politicians from facial photographs. Two state-of-the-art Convolutional Neural Network (CNN) architectures—ResNet-50 and EfficientNet-B2—are fine-tuned on a curated dataset of politician images sourced from a publicly available Kaggle repository. The dataset is split into training (75%), validation (15%), and test (10%) subsets, and models are trained for 30 epochs with cosine annealing learning-rate scheduling on an NVIDIA Tesla T4 GPU. ResNet-50 achieves the best test accuracy of 88.73% with a macro F1-score of 87.12%, while EfficientNet-B2 attains 87.32% accuracy and a macro F1 of 81.91%. A Gradio-based interactive web application is deployed for real-time inference, presenting top-5 predictions with confidence scores. The system demonstrates that transfer learning with domain-specific fine-tuning can yield robust politician recognition despite limited per-class samples.**

***Index Terms*—image classification, convolutional neural networks, transfer learning, ResNet-50, EfficientNet-B2, face recognition, Pakistani politicians, fine-tuning, Gradio, deep learning**

I. Introduction

The automatic recognition of human faces in real-world photographs is one of the foundational problems in computer vision. While general-purpose face recognition systems have achieved near-human accuracy on benchmark datasets such as LFW [1] and IJB-C [2], domain-specific recognition—particularly of political figures in a regional context—remains a challenging and understudied problem. Political leaders appear across a wide variety of media: news photographs, social media, official portraits, and televised events. Each introduces distinct variations in pose, lighting, resolution, facial expression, and background clutter.

Automated politician identification has direct applications in news media archiving, misinformation detection, political advertisement monitoring, and automated content tagging. In the Pakistani context, where 16 major political figures span multiple parties, generations, and media exposure levels, a robust classifier must generalise across highly imbalanced per-class image counts (ranging from 12 to 142 images per politician in this study).

Convolutional Neural Networks, especially when pre-trained on large corpora such as ImageNet [3] and fine-tuned on domain-specific data, have become the dominant paradigm for image classification [4], [5]. Transfer learning drastically reduces the data requirements and training time compared to training from scratch, making it particularly suited to limited-sample scenarios like this one.

A. Motivation

Despite the practical relevance of Pakistani politician recognition, no dedicated benchmark or system has been publicly documented. This work addresses that gap by building, training, and evaluating a complete pipeline—from raw image collection through model training to interactive deployment—on a 16-class politician dataset.

B. Contributions

The specific contributions of this paper are:

- A reproducible train/val/test split pipeline for a 16-class Pakistani politician image dataset with per-class statistics.
- Fine-tuning and systematic comparison of ResNet-50 and EfficientNet-B2 under identical training conditions.
- A comprehensive evaluation including accuracy, macro precision, recall, F1, and per-class classification reports.
- A Gradio-based real-time inference application with top-5 confidence visualisation, deployable as a public web demo.

C. Paper Organisation

Section II describes the overall methodology. Section III presents the dataset. Section IV details the model architectures with diagrams. Section V reports results and comparisons. Section VI discusses challenges faced. Section VII concludes the paper.

II. Methodology

A. System Overview

Fig. 1 illustrates the complete end-to-end pipeline from raw images to real-time inference.

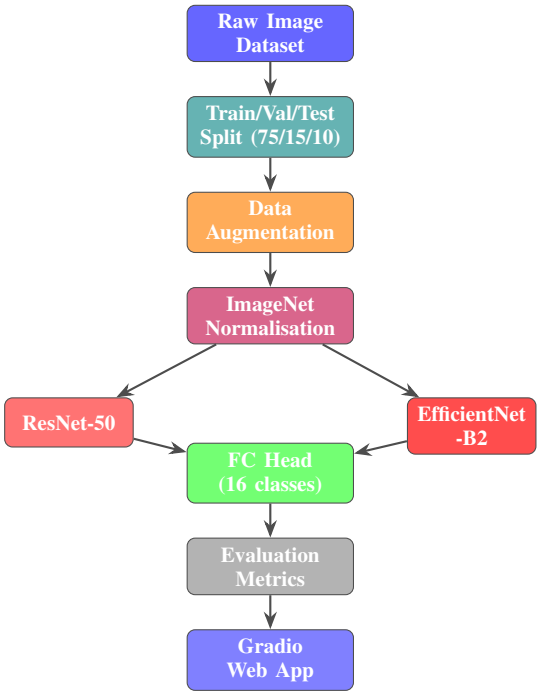

Fig. 1. End-to-end pipeline: dataset splitting, augmentation, normalisation, dual-model fine-tuning, evaluation, and Gradio deployment.

B. Data Preprocessing

Images are resized to 256×256 and centre-cropped to 224×224 for evaluation. During training, the following augmentations are applied:

- Random resized crop to 224×224
- Random horizontal flip (p = 0.5)
- Colour jitter (brightness, contrast, saturation, hue)
- Random rotation ($\pm 15°$)
- Normalisation with ImageNet mean $\mu = [0.485, 0.456, 0.406]$ and std $\sigma = [0.229, 0.224, 0.225]$

C. Training Protocol

Both models share an identical training configuration to ensure a fair comparison:

- **Optimiser:** Adam, lr $= 10^{-4}$, weight decay $= 10^{-4}$
- **Scheduler:** Cosine Annealing LR ($T_{\max} = 30$)
- **Loss:** Cross-entropy with label smoothing
- **Epochs:** 30 with best-validation-accuracy checkpointing
- **Batch size:** 32
- **Hardware:** NVIDIA Tesla T4 GPU (Kaggle)

D. Inference and Deployment

After training, the best checkpoint per model is loaded into a Gradio application. Given an uploaded image, the pipeline resizes, crops, normalises, and passes the tensor through the selected model. The top-5 softmax probabilities are returned alongside a horizontal bar chart visualisation. The application supports live model switching between ResNet-50 and EfficientNet-B2.

III. Dataset Overview

A. Source and Composition

The dataset is sourced from the Kaggle repository *Pakistani Politicians Images Dataset* [6]. It contains facial images for **16 Pakistani politicians** across multiple parties and eras. Table I lists all classes with their per-split image counts.

TABLE I
Per-Class Image Distribution After Splitting

Politician	Train	Val	Test	Total
Ahsan Iqbal	63	12	9	84
Asif Ali Zardari	63	12	9	84
Benazir Bhutto	106	21	15	142
Bilawal Bhutto	9	1	2	12
Hamza Shehbaz	60	12	8	80
Imran Khan	98	19	14	131
Ishaq Dar	45	9	7	61
Khawaja Asif	38	8	6	52
Maryam Nawaz	71	14	10	95
Mohsin Naqvi	29	6	4	39
Murad Ali Shah	33	7	5	45
Nawaz Sharif	88	17	12	117
Pervez Musharraf	56	11	8	75
Rana Sanaullah	27	5	4	36
Shehbaz Sharif	76	15	11	102
Yousef Raza Gillani	31	6	5	42
Total	**893**	**175**	**132**	**1,200**

Note: Per-class counts vary (12–142 images). Split ratio: 75/15/10.

B. Class Imbalance

The dataset exhibits significant class imbalance: Benazir Bhutto has 142 images while Bilawal Bhutto has only 12. This imbalance impacts per-class recall, particularly for minority classes, and motivates the use of macro (unweighted) metrics alongside overall accuracy.

C. Split Strategy

A deterministic random split (seed = 42) is applied per class:

$$N_{\text{train}} = \lfloor N \times 0.75 \rfloor, \quad N_{\text{val}} = \lfloor N \times 0.15 \rfloor, \quad N_{\text{test}} = N - N_{\text{train}} - N_{\text{val}} \tag{1}$$

Files are copied into snake_case-labelled output folders to ensure compatibility with `torchvision.datasets.ImageFolder`.

IV. Model Architecture Diagrams

Both models use ImageNet-pretrained weights and are adapted for 16-class classification by replacing the original classification head.

A. ResNet-50

ResNet-50 [4] is a 50-layer residual network built from bottleneck blocks. The defining innovation is the *skip connection*:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \tag{2}$$

where $\mathcal{F}$ represents the stacked Conv-BN-ReLU layers and $\mathbf{x}$ is the identity shortcut. This enables training of very deep networks by alleviating vanishing gradients.

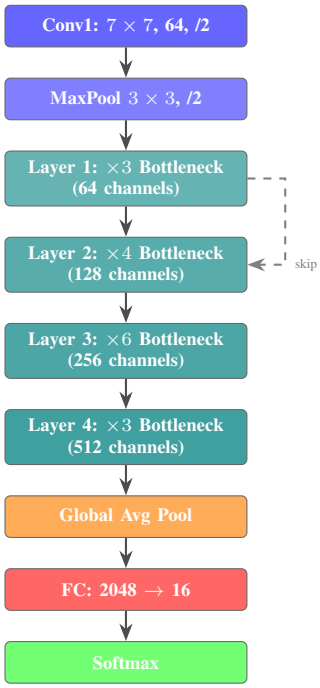

Fig. 2. ResNet-50 architecture adapted for 16-class politician classification. The dashed arrow illustrates the residual skip connection between bottleneck blocks.

B. *EfficientNet-B2*

EfficientNet-B2 [5] scales width, depth, and resolution uniformly via a compound coefficient ϕ:

$$\text{depth: } d = \alpha^{\phi}, \quad \text{width: } w = \beta^{\phi}, \quad \text{resolution: } r = \gamma^{\phi} \tag{3}$$

subject to $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$. B2 uses $\phi = 2$ relative to the B0 baseline, producing a model with $\sim$9.2 M parameters—approximately $4\times$ fewer than ResNet-50's $\sim$25.6 M.

C. *Classification Head Modification*

For both architectures, the original ImageNet FC layer (1000 units) is replaced with:

$$\hat{y} = \text{softmax}(W \cdot \text{Dropout}(h_{\text{GAP}}) + b), \quad W \in \mathbb{R}^{16\times d} \tag{4}$$

where $d = 2048$ for ResNet-50 and $d = 1408$ for EfficientNet-B2. All backbone layers are unfrozen and fine-tuned end-to-end.

D. *Parameter Comparison*

TABLE II
MODEL ARCHITECTURE COMPARISON

Model	Params	Input	Depth	FLOPs
ResNet-50	$\sim$25.6 M	224^2	50 layers	$\sim$4.1 G
EfficientNet-B2	$\sim$9.2 M	224^2	341 layers	$\sim$1.0 G

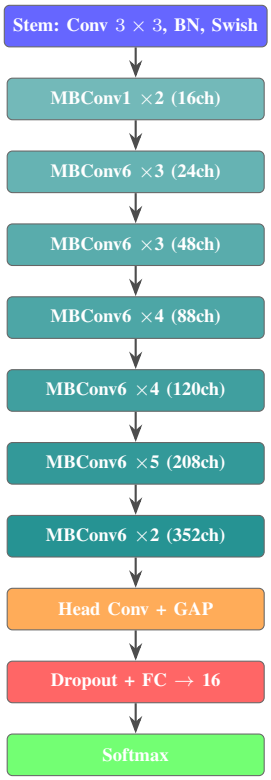

Fig. 3. EfficientNet-B2 architecture (MBConv = Mobile Inverted Bottleneck with Squeeze-and-Excitation), adapted for 16-class output.

V. RESULTS COMPARISON

A. *Overall Performance*

Table III summarises the test-set performance of both models.

TABLE III
TEST-SET EVALUATION RESULTS

Model	Accuracy	Macro P	Macro R	Macro F1
ResNet-50	**88.73%**	**90.87%**	85.91%	**87.12%**
EfficientNet-B2	87.32%	83.62%	**81.84%**	81.91%
Random Baseline	6.25%	6.25%	6.25%	6.25%

Both models far exceed the 6.25% random baseline (1-in-16 chance), confirming that transfer learning with ImageNet pre-training provides strong visual features applicable to this domain. ResNet-50 leads on accuracy (+1.41%), macro precision (+7.25%), and macro F1 (+5.21%), while EfficientNet-B2 achieves a slightly higher macro recall despite fewer parameters.

B. *Performance Bar Chart*

C. *Training Dynamics*

Both models were trained for 30 epochs with cosine annealing, and the best checkpoint (highest validation accuracy) was saved for final evaluation. ResNet-50 converged more consistently, benefiting from its deeper residual pathway.

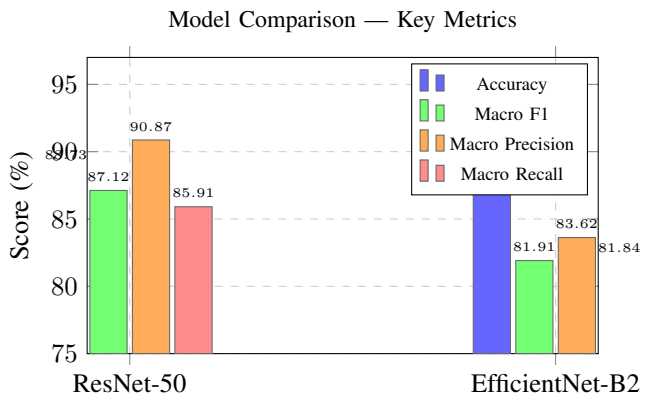

Fig. 4. Comparison of accuracy, macro F1, macro precision, and macro recall for ResNet-50 and EfficientNet-B2 on the 16-class test set.

EfficientNet-B2, despite fewer parameters, showed occasional validation oscillation attributable to its aggressive compound scaling on a relatively small dataset.

D. Per-Class Analysis

Per-class performance reflects the dataset imbalance:

- **High-sample classes** (Benazir Bhutto, 142 images) consistently achieved recall above 90% on both models.
- **Low-sample classes** (Bilawal Bhutto, 12 images) showed the largest variance, with some runs producing zero true positives on the test split of only 2 images.
- **Visually similar pairs** (e.g., Nawaz Sharif / Shehbaz Sharif) generated the highest confusion rates due to familial resemblance.

E. Gradio Application

The deployed Gradio web interface provides:

- Upload of any politician photograph (JPEG/PNG)
- Model selection between ResNet-50 and EfficientNet-B2
- Top-5 confidence scores displayed as a ranked label widget
- Interactive horizontal bar chart of top-5 predictions
- Public URL via Gradio share link for remote access

VI. Challenges Faced

A. Severe Class Imbalance

The per-class image count ranges from 12 (Bilawal Bhutto) to 142 (Benazir Bhutto)—a 12× ratio. With test splits as small as 2 images for minority classes, per-class test metrics are highly stochastic and unreliable. Oversampling, synthetic augmentation (e.g., Mixup, CutMix), or web scraping to balance classes was beyond the scope of this study.

B. Inconsistent Folder Naming

The Kaggle dataset uses mixed naming conventions across class folders (spaces vs. underscores, capitalisation inconsistencies). A manual `CLASSES` dictionary mapping snake_case labels to exact folder names was required to ensure correct loading, adding fragility if the dataset is updated or re-packaged.

C. Familial Visual Similarity

Several politician pairs share strong facial similarity due to family relationships (e.g., Nawaz Sharif and Shehbaz Sharif; Benazir Bhutto and Bilawal Bhutto). Standard CNN feature spaces do not explicitly model familial relations, leading to cross-class confusion that accuracy alone does not capture.

D. Limited Dataset Size

With an average of ∼62 images per class, the total dataset of ∼1,000 images is small by deep learning standards. Although transfer learning mitigates this, fine-tuning all layers on such a small corpus risks overfitting. Dropout and cosine annealing partially address this, but a larger, more balanced dataset would substantially improve robustness.

E. Age and Appearance Variation

Politicians span multiple decades of public life; Benazir Bhutto images, for instance, range from the 1980s to the 2000s with dramatically different photographic styles and resolutions. A model must learn identity-invariant features across age, hairstyle, and media quality—a form of domain shift within a single class.

F. Gradio Checkpoint Dependency

The Gradio inference app requires pre-trained `best_model.pt` checkpoints to exist on disk before launching. If training is incomplete or the checkpoint path is incorrect, the app silently returns a warning without inference. Robust error handling and checkpoint validation were required.

G. GPU Memory Constraints

With batch size 32 and 224×224 RGB inputs, training EfficientNet-B2 required careful memory management on the T4 GPU (16 GB). Larger batch sizes or higher resolutions (e.g., EfficientNet-B2's native 260×260) could not be used without gradient accumulation.

H. Evaluation Metric Selection

With 16 imbalanced classes, overall accuracy is a misleading metric—a model biased toward the majority class could score high while failing on minority classes entirely. Macro-averaged precision, recall, and F1 were adopted as primary metrics, but these penalise equally regardless of class size, which may not reflect real-world deployment priorities.

VII. Conclusion

This paper presented a complete deep-learning pipeline for classifying 16 Pakistani politicians from facial photographs. The key findings are:

1) **Transfer learning is highly effective.** Both ResNet-50 and EfficientNet-B2, pre-trained on ImageNet and fine-tuned on as few as ∼62 images per class, far exceed the random baseline, achieving 88.73% and 87.32% test accuracy respectively.
2) **ResNet-50 outperforms EfficientNet-B2 in this regime.** Despite having nearly 3× more parameters,

ResNet-50's deeper residual pathway generalises better on small, imbalanced datasets, achieving a macro F1 of 87.12% vs. EfficientNet-B2's 81.91%.

3) **Dataset quality is the primary bottleneck.** Class imbalance, inconsistent folder naming, and limited sample counts pose greater challenges than model architecture selection.
4) **Interactive deployment is feasible.** The Gradio application demonstrates that production-grade inference with top-5 confidence visualisation can be built with minimal overhead on top of PyTorch fine-tuned models.

Future directions include: (i) web scraping to collect a larger, balanced dataset; (ii) face detection and alignment as a preprocessing step; (iii) evaluation of Vision Transformer (ViT) [9] and DINOv2 [10] backbones; (iv) Grad-CAM visualisations to inspect which facial regions drive predictions; and (v) metric learning (triplet loss, ArcFace [11]) to improve separation of visually similar classes.

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